Mittal Variety Store: "Driving B2C Success Through Data Driven Solutions Stock Optimization and Sales growth".

A Mid Term report for the BDM capstone Project

Submitted by

Name: **Somiya**

Roll number: **24F2002516**

IITM Online BS Degree Program,

Indian Institute of Technology, Madras,

Tamil Nadu, India, 60003

Content

1.Executive Summary	**3**
2.Proof of Originality	**3-4**
3.Metadata	**4-7**
4. Detailed Explaination of Analysis	**7-8**
4.Result & Findings	**8-12**
5.Conclusion	**12**

Declaration Statement

I am working on a Project Title "**Mittal Variety Store: "Driving B2C Success Through Data Driven Solutions Stock Optimization and Sales growth".** I extend my appreciation to **Mittal variety Store ,** for providing the necessary resources that enabled me to conduct my project. I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability. Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures. I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report. I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority. I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:

Name: Somiya

Date: Dec 26,2025

1.Executive summary

Mittal Variety Store, a local grocery shop located in Jaito, Punjab, is experiencing the numerous operational challenges in the tracking of inventory pricing and credit handling. Business comes under the category of Business-to-consumer(B2C) outlet for more than five years, the store provides daily-routine groceries which is essential for everyone including biscuits, namkeen, spices, and household Products. Moreover, the shop owner follows the traditional manual methods which resulted in frequent stock-outs overstocking inefficient pricing and unstructured credit records, following to the revenue losses and decline in a profitability. To tackle these problems, The Business data management project aims to execute data-driven solutions by analysing and gathering sales, stock and credit transaction data from the store. The data has been collected from the store's manuals records and structured into a dataset which is suitable for analysis. Initial Cleaning of the data was carried out to remove consistencies and prepare it for further analysing. Early Observations indicate the presence of slow moving and expired goods, fluctuations or irregularities in sales trends and late payments from customers affecting Cash Flow. Analytical techniques such as ABC analysis and stock analysis, Credit vs Non-Credit analysis receivables Tracking, category Wise Revenue Analysis and also the Cash flow Analysis and data visualization are being implemented to identify demanding items, Optimize Stock levels and establish credit control systems. By leveraging these findings, the projects aims to provide the store owner with practical analysis that can make improvement in the profitability, inventory management and customer satisfaction.

2. Proof of Originality:

To ensure compliance with the originality requirements, all supporting evidence has been compiled and submitted in a single Google Drive folder for verification. The submitted materials include the following:

1. **Video Interview:**

   A recorded interview with Mr. Rajinder Kumar, owner of Mittal variety Store, discussing the shop's background, business operations, challenges, and expectations from this project. The interview was conducted in Hindi, and an English transcript of the conversation has been provided.

2.Photographs:

Images of the business premises, including the shop setup, operational areas, and relevant visuals reflecting day-to-day business activities. Photographs of the shop owner have also been included.

3. Official Consent Letter:

A signed consent letter from the business owner confirming my involvement in the project and verifying the authenticity of the collected data.

Google Drive Link (All Materials): Copy paste it on google chrome
https://drive.google.com/drive/folders/1GF89G1hIVBPzCsGDt8DKFdfyOoRr21pi

3. Metadata and Descriptive Statistics:

Metadata:

● Dataset Overview:

This dataset contains detailed records related to grocery sales, stock movement, and payment modes for the period from January 2025 to June 2025. It includes key information such as total units sold, selling prices, closing stock, profit margins, and credit sales , offering a comprehensive view of the store's retail performance. The data has been systematically collected from **Mittal Variety Store** sales bills, inventory registers, and financial transaction records, and has been carefully compiled and organized to support accurate analysis and interpretation.

● Dataset Size:

Dataset 1: Grocery Sales Dataset

1. **Number of Records (Rows):** 4065
2. **Number of Attributes (Columns):** 12
3. **Categorical Columns:** Date, Product_SKU ,Product_Name ,Category, Sub_Category , Payment_Mode ,Is_Credit , Credit_Sales.
4. **Numerical Columns:** Quantity_Sold , Selling_price , Revenue, Profit.
5. **Purpose of Dataset:**This dataset is used to analyze sales performance, customer purchasing behavior, revenue generation, payment trends, and profitability of the grocery store. It helps in identifying high-performing products, preferred payment methods, and credit-based sales patterns.

Dataset 2: Grocery Stocks Dataset

1. **Number of Records (Rows):**6155

2. **Number Of Attributes (Columns):**6
3. **Categorical Columns:** Date , Product_SKU , Product_Name.
4. **Numerical Columns:**Opening_Stock , Stock_Received , Closing_Stock.
5. **Purpose of Dataset:** This dataset is used to analyze inventory movement, stock availability, and stock control efficiency. It helps in identifying stock shortages, overstocking issues, and supports effective inventory planning and reorder decisions.

- **3. Attributes:**

Dataset 1: Grocery Sales Dataset

1. **Product_Name (String):** Used to identify and compare product-wise sales.
2. **Category (String):** Used for category-wise sales analysis.
3. **Quantity_Sold (Integer):** Used to measure product demand and sales volume.
4. **Selling_Price (Float):** Used to calculate revenue.
5. **Revenue (Float):** Used to evaluate total sales performance.
6. **Profit (Float):** Used to analyse profitability of products.
7. **Payment_Mode (String):** Used to study customer payment behaviour.

Dataset 2: Grocery Stock Dataset

1. **Product_Name (String):** Used to identify stock items.
2. **Opening_Stock (Integer):** Used to analyse starting inventory level.
3. **Stock_Received (Integer):** Used to track incoming inventory.
4. **Closing_Stock (Integer):** Used to analyse remaining stock and stock-outs.

Link to the Dataset: [Dataset]

Working Analysis with Dataset: [Analysis of data]

Descriptive Statistics:

1. Dataset Overview:

The dataset consists of grocery sales and stock records collected over a specific period. It contains multiple attributes related to product sales, inventory levels, pricing, revenue, and profit. The dataset includes both categorical and numerical variables, enabling a detailed analysis of sales performance, customer purchase behaviour, and inventory management. This dataset provides a comprehensive view of operational efficiency and business performance in a retail grocery environment.

2. Measures of Central Tendency:

Mean (Average): The mean values are used to understand the overall performance of the grocery store.

Measure	Quantity Sold	Selling Price (₹)	Revenue (₹)	Profit (₹)
Mean (Average)	1.86	74.01	133.10	18.69
Median	2	60	90	12
Minimum	1	25	35	4
Maximum	3	320	960	120

The average quantity sold per transaction is 1.86 units, indicating a moderate sales volume across transactions. The average selling price is ₹74.01, which reflects the general pricing level of products in the store. On average, the revenue generated per transaction is ₹133.10, showing steady overall sales performance. The average profit earned per entry is ₹18.69, indicating consistent profitability. From the inventory perspective, the average opening stock is 47.32 units, while the average stock received is 0.94 units, suggesting limited daily replenishment. The average closing stock stands at 47.04 units, reflecting a balanced flow of inventory without excessive accumulation or shortage.

Median: The median values help in understanding the typical behaviour of sales and stock data:

Measure	Opening Stock	Stock Received	Closing Stock
Mean (Average)	47.32	0.94	47.04
Median	43	0	43
Minimum	17	0	17
Maximum	120	50	120

The median quantity sold is 2 units, which shows that most customer transactions involve small quantities. The median selling price is ₹60, highlighting a focus on affordable products. The

median revenue is ₹90, representing a common transaction value observed in the dataset. The median profit is ₹12, indicating consistent earnings per sale. In terms of inventory, the median opening stock is 43 units, and the median closing stock is also 43 units, suggesting stable stock levels and minimal fluctuations in inventory movement

3. Frequency Analysis:

The frequency analysis of product categories reveals that Staples have the highest number of transactions (1352), followed by Personal Care products (1065). Packaged Food items account for 568 transactions, while Household Items contribute 396 entries. Beverages and Dairy Products show relatively lower frequencies with 373 and 311 transactions respectively. This distribution indicates higher customer demand for essential and daily-use products compared to other categories.

4.Detailed Explanation of Analysis Process/Method:

The analysis was first carried out for Mittal Variety Store with the aim of understanding real business operations and customer purchasing behaviour. The dataset includes sales and stock details of grocery products collected over a specific period and was analysed using structured data analysis techniques. The main purpose of this analysis was to evaluate business performance, study inventory efficiency, and identify customer buying patterns. By examining sales quantity, revenue, profit, and stock movement, the analysis helps in understanding demand trends and supports better decision-making for inventory and sales management.

1. Data Collection and Preparation

- The dataset was created to represent practical grocery store operations, including sales transactions and inventory movement.

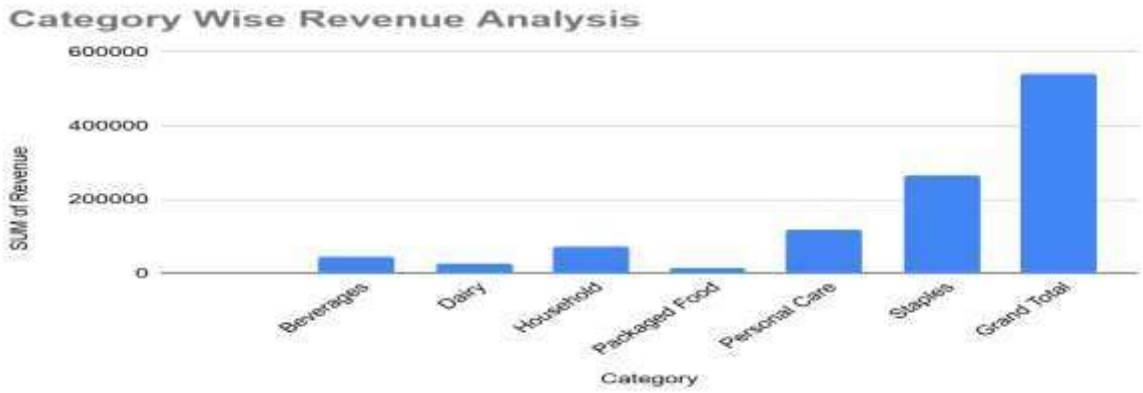
  Data cleaning and preprocessing were carried out using **Microsoft Excel and Python** (Pandas) to ensure accuracy, consistency, and proper formatting of records.
- Irrelevant and duplicate fields were removed to maintain clarity and improve the quality of analysis.
- The dataset was organized systematically to support sales analysis, inventory management, and profitability evaluation.
- The final dataset contained the following important attributes: **Product_Name, Category, Quantity_Sold, Selling_Price, Revenue, Profit, Payment_Mode, Opening_Stock, Stock_Received, and Closing_Stock.**.

2. Descriptive Statistical Analysis:

- Descriptive statistics had generated using Python's describe() function to summarize and understand the overall behaviour of the dataset.
- The analysis focused on important numerical variables like **quantity sold, selling price, revenue, profit, and stock levels** to analyse the business performance.
- Measures such as **mean, median, and standard deviation** were used to examine the central tendency and variation in sales and inventory data.
- **Collab Notebook :-click here**

5. Results and findings:

- **Category-wise Revenue analysis:**

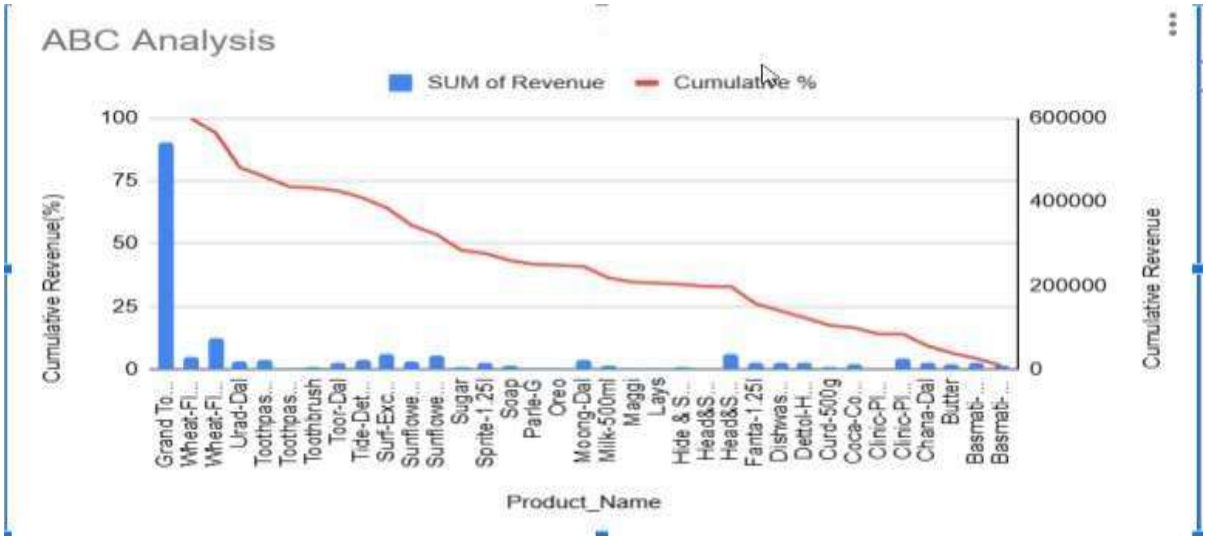


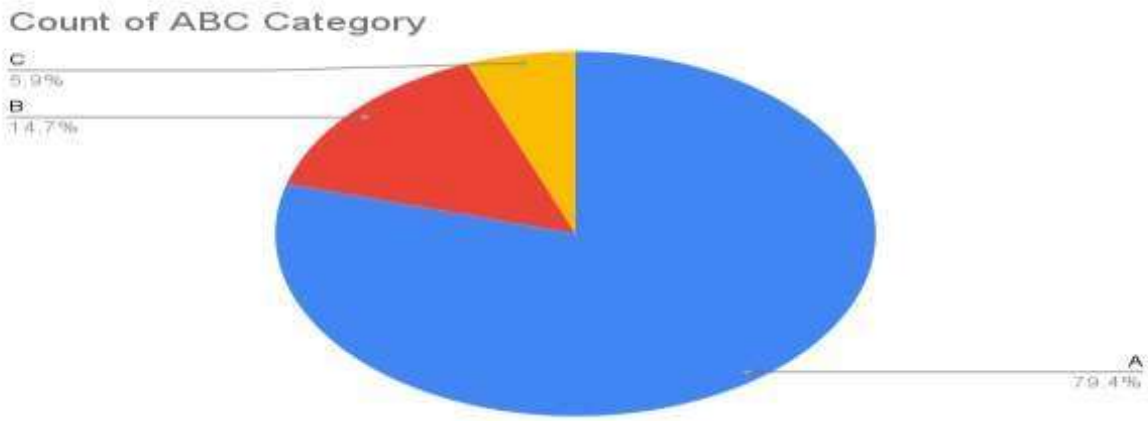
Observation from the Graph

The chart represents the total revenue generated by different product categories, highlighting their contribution to the overall sales of the store.

- Staples clearly dominate the revenue share, indicating that essential grocery items contribute the most to total sales.
- Personal Care and Household categories follow next, forming the second-highest revenue group due to regular consumer demand.
- Beverages and Packaged Food contribute a moderate portion of revenue, reflecting occasional or selective purchases by customers.

- **ABC ANALYSIS:**





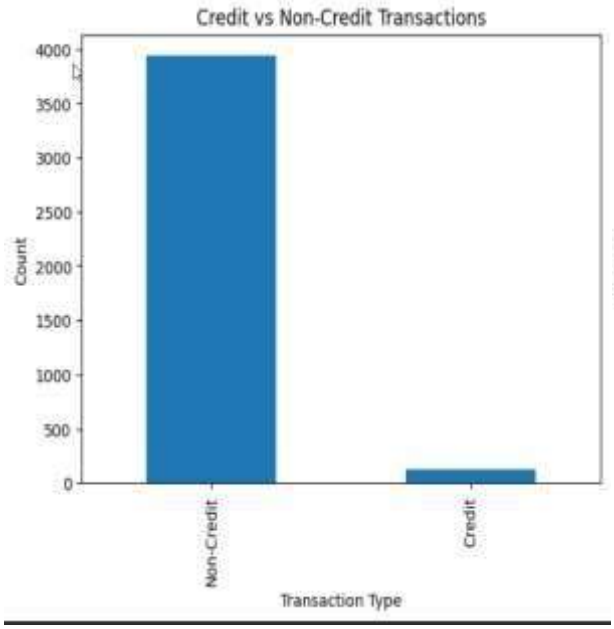
Observations from ABC Analysis Graph:

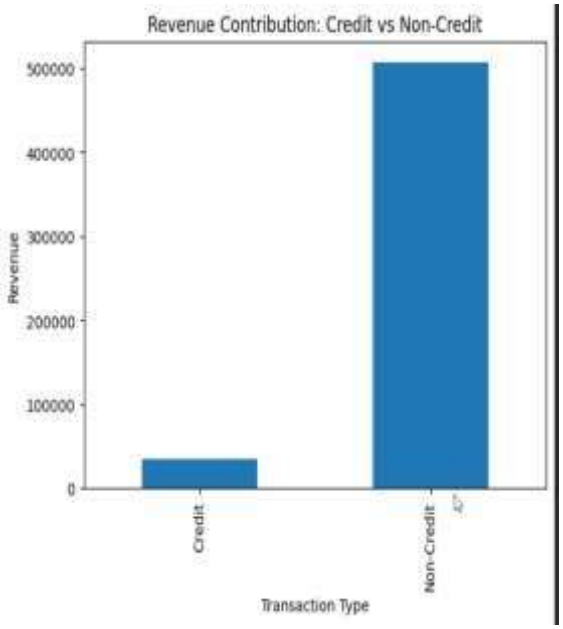
- The graph represents ABC classification based on cumulative revenue contribution of products.

  Category A products (left side of the graph) generate the highest revenue and include high-value and fast-moving items such as wheat flour, edible oil, and branded grocery items. These products require strict inventory control and regular monitoring.

- Category B products contribute moderately to total revenue and show a gradual increase in the cumulative curve. These items have stable demand and require periodic review.

- Category C products form the largest group in terms of quantity but contribute the least to total revenue, as seen from the flattening of the curve toward the end. These items are low-value or slow-moving products.
- The curve clearly follows the **Pareto Principle (80–20 rule)**, where a small percentage of items account for the majority of revenue.
- **Credit Analysis:**

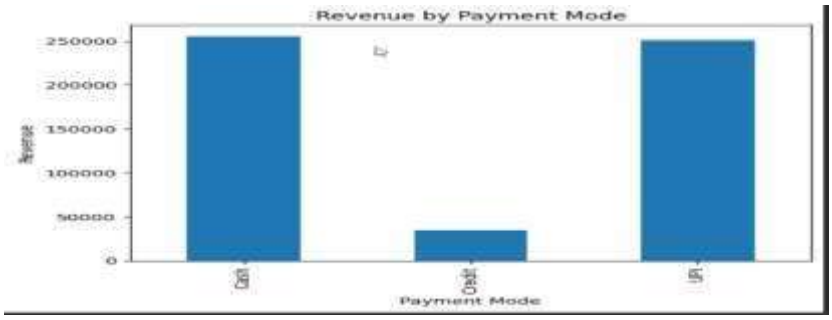




Observations from the Credit Analysis Graph

- The graph compares Credit and Non-Credit transactions in terms of both transaction count and revenue contribution.
- It is clearly observed that non-credit transactions dominate in number, indicating that most customers prefer making immediate payments rather than purchasing on credit.
- The revenue generated from non-credit transactions is significantly higher than that from credit transactions, showing strong cash flow for the business.

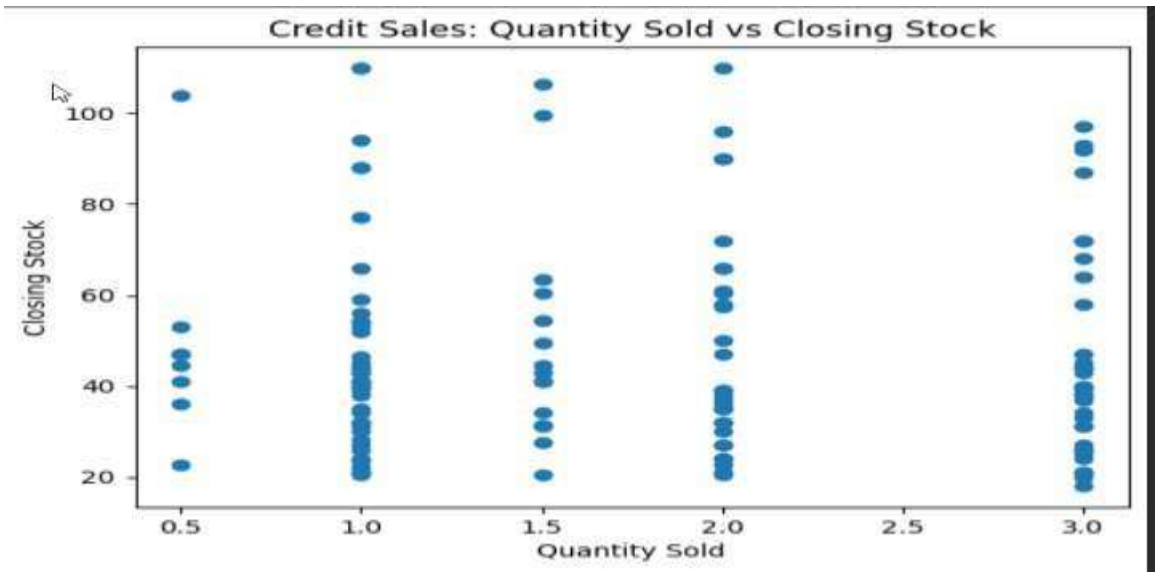
  Credit transactions form only a small portion of total sales, suggesting limited dependency on credit-based purchases..
- **Cash flow:**



Observations from this graph

- Cash and UPI payments generate the highest revenue, indicating that most customers prefer immediate payment methods.
- Credit transactions contribute significantly less revenue, showing limited reliance on credit purchases.
- The overall trend suggests strong cash flow and low credit dependency, which is positive for business stability.

Stock Analysis:



Observations for this graph:

- The scatter plot shows an **inverse relationship** between quantity sold and closing stock as sales increase, closing stock generally decreases.
- Most credit sales occur when **quantity sold is between 1 and 3 units**, indicating small volume credit purchases.
- Higher closing stock values are observed when **sales quantity is low**, suggesting slow movement of certain items.

- Some products still show high closing stock even after sales, indicating **overstocking or low demand**. Collab Notebook: Graph Analysis

Recommendations:

- According to the ABC analysis, more focus should be given to **Category A products** as they generate the highest revenue. These items should always be kept in sufficient stock.
- **Category C products** have lower sales, so their stock should be limited to avoid blocking unnecessary capital.
- The credit analysis shows that **cash and UPI payments are used more frequently**, so these payment modes should be encouraged.
- Credit sales should be kept under control to avoid payment delays and financial losses.

6. Conclusion:

The Mittal Variety Store investigation reveals a number of significant facts about client purchase patterns, inventory control, and sales success. Essential grocery items make up the largest portion of total income, according to the survey, demonstrating consistent demand for everyday goods. While certain products exhibit little mobility and call for greater stock planning or promotional efforts, some categories do better than others. Additionally, the data shows that cash and digital payments account for the majority of transactions, indicating a strong cash flow and minimal reliance on credit. According to inventory trends, efficiency can be further increased by having better control over both slow-moving and fast-moving products. Overall, the store may increase profitability and guarantee more seamless business operations by concentrating on high-performing products, managing inventory more efficiently, and matching sales tactics with consumer behaviour.